

Stagehand Phase Evidence

Current checkpoint: public Site v94, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Source HEAD: c28911081af6a7f611bb684cf84235637ece04e7

Deployed runtime: 2312388c03d4f245378fe9ec667cd246852859c3

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 94

Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Asset treatment choices reach canvas, thumbnails, Preview, and WebM; new assets also receive valid style defaults immediately.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Human audio cue Start/End editors share bounded timing updates with the agent cue editor.
- Animate and Storyboard are the only top-level editor modes; Preview is a compact review action and overflow actions keep infrastructure chrome out of the workspace.
- The timeline defaults to semantic camera, pose, prop, dialogue, music, and SFX event bands; Show details restores raw keyframe controls.
- Responsive project and Inspector drawers preserve a readable shell at phone widths.
- At split-browser widths up to 1200 px, the Inspector is closed by default and opens as an overlay drawer while the stage and project rail retain working width.
- A Sol-powered Codex run used the public page's live WebMCP tools to apply and validate an Alice/Bob coupon gag without a separate MCP connection or TOML change.

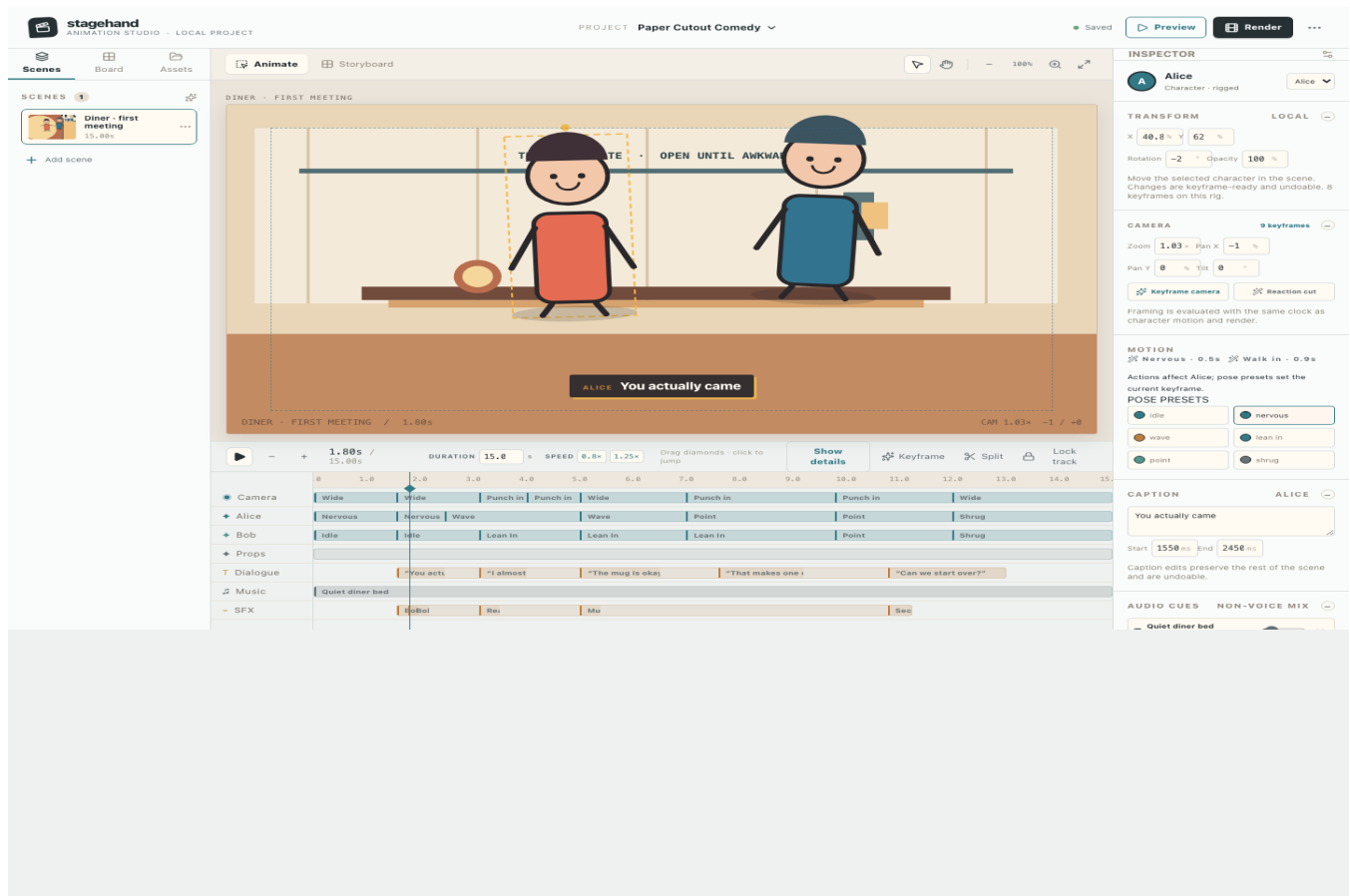
Verification

Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
npm run smoke:native
```

Smoke passed: 52 tools, 39 guarded mutations, a 15-second six-beat starter, synchronized timing, asset treatments, import/recovery including empty collections, Preview, valid PNG/WebM signatures, native WebMCP registration, and zero errors.

Animate editing workspace



Hosted Playwright capture at 1440 x 960. The focused editor keeps artwork dominant, with semantic event bands and raw-keyframe details available on demand.

Assets style editor

The screenshot shows the Stagehand Animation Studio interface. The main workspace displays a scene titled "DINER - FIRST MEETING" with two characters, Alice and Bob, in a diner setting. Alice is a red character and Bob is a blue character. The interface includes a left sidebar with asset lists, a top navigation bar, and a right sidebar with the Asset Inspector. The Asset Inspector shows details for "Alice - rigged" and "Bob - rigged" assets, including their placement and style settings. The bottom timeline shows a duration of 15.0 seconds and a speed of 0.8x.

Assets

- Alice - rigged
Rigged character - starter - ...
HERO - PAPER
CORAL - DINER TEAL
- Bob - rigged
Rigged character - starter - ...
HERO - PAPER
DINER TEAL - WARM PAPER
- Diner background
Background - starter - ON STAGE
ENVIRONMENT - PAPER
WARM PAPER - DINER TEAL
- Coffee mug
Prop - starter - LIBRARY
ACCENT - FLAT-COLOR
MUSTARD - AMBER

ADD PLACEHOLDER

- + Rigged character
- + Background
- + Prop
- + Audio
- Import character art
- Import pose sheet - auto
- Import background
- Import prop

Import character art to bind it to the selected rig; poses and keyframes stay editable. Pose-sheet imports auto-detect a wide four-column image and fall back to single art when needed.

ASSET INSPECTOR

Alice - rigged
RIGGED CHARACTER - STARTER

PLACEMENT UNBOUND
Available for rig binding.

STYLE PAPER
Hero Clear Silhouette

[Edit style](#)

Asset identity and art direction live here. Motion and rig controls return when you select Scenes.

Timeline

1.00s / 15.00s DURATION 15.0 s SPEED 0.8x 1.25x

Camera: Wide, Wide, Punch in, Punch in, Wide, Punch in, Punch in, Wide

Alice: Nervous, Nervous, Wave, Wave, Point, Point, Shrug

Bob: Idle, Idle, Lean in, Lean in, Lean in, Point, Shrug

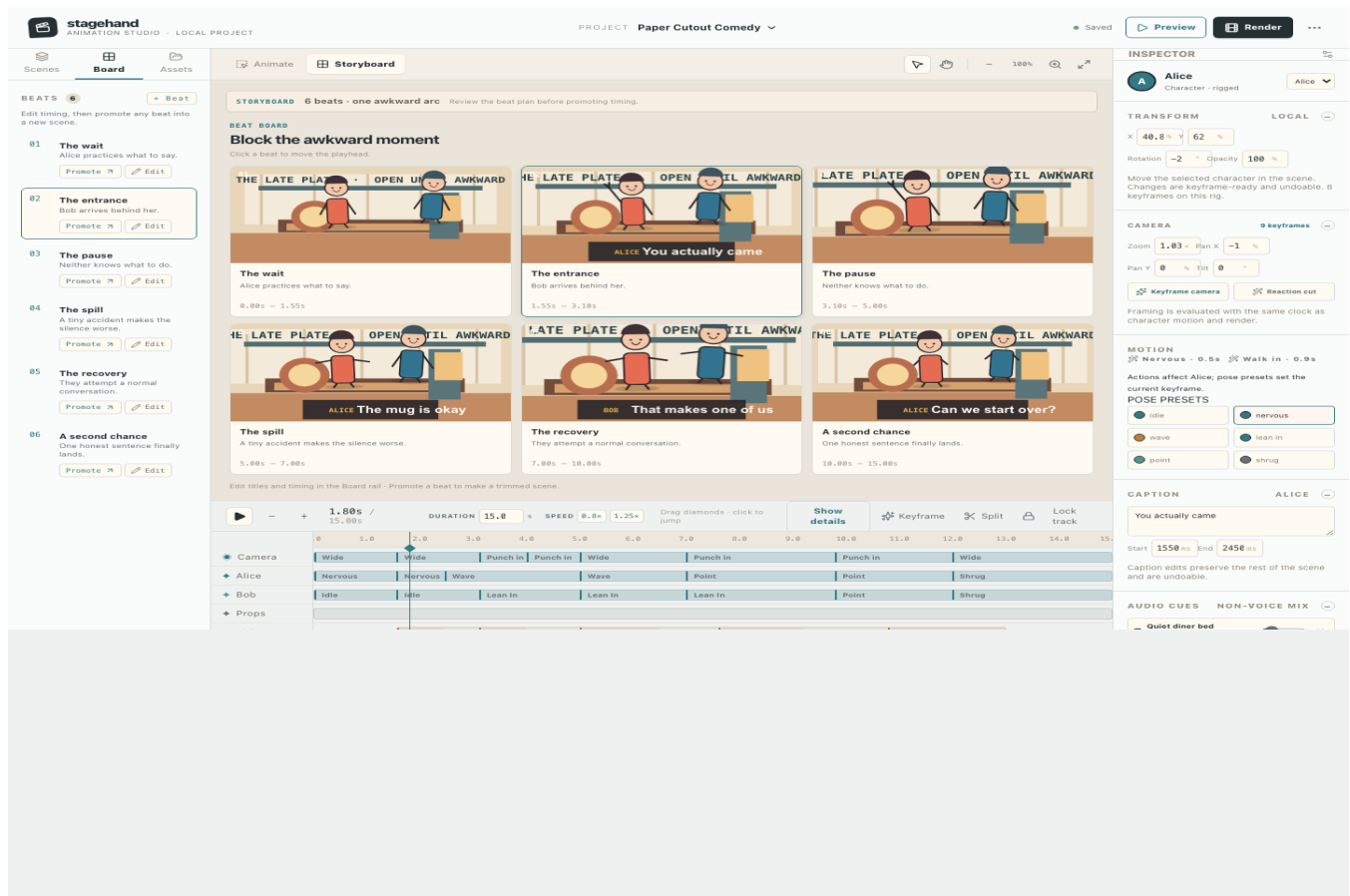
Dialogue: "You acti", "I almost", "The mug is okay", "That makes one", "Can we start over?"

Music: Quiet diner bed

SFX: SoBol, Rei, Mu, Sec

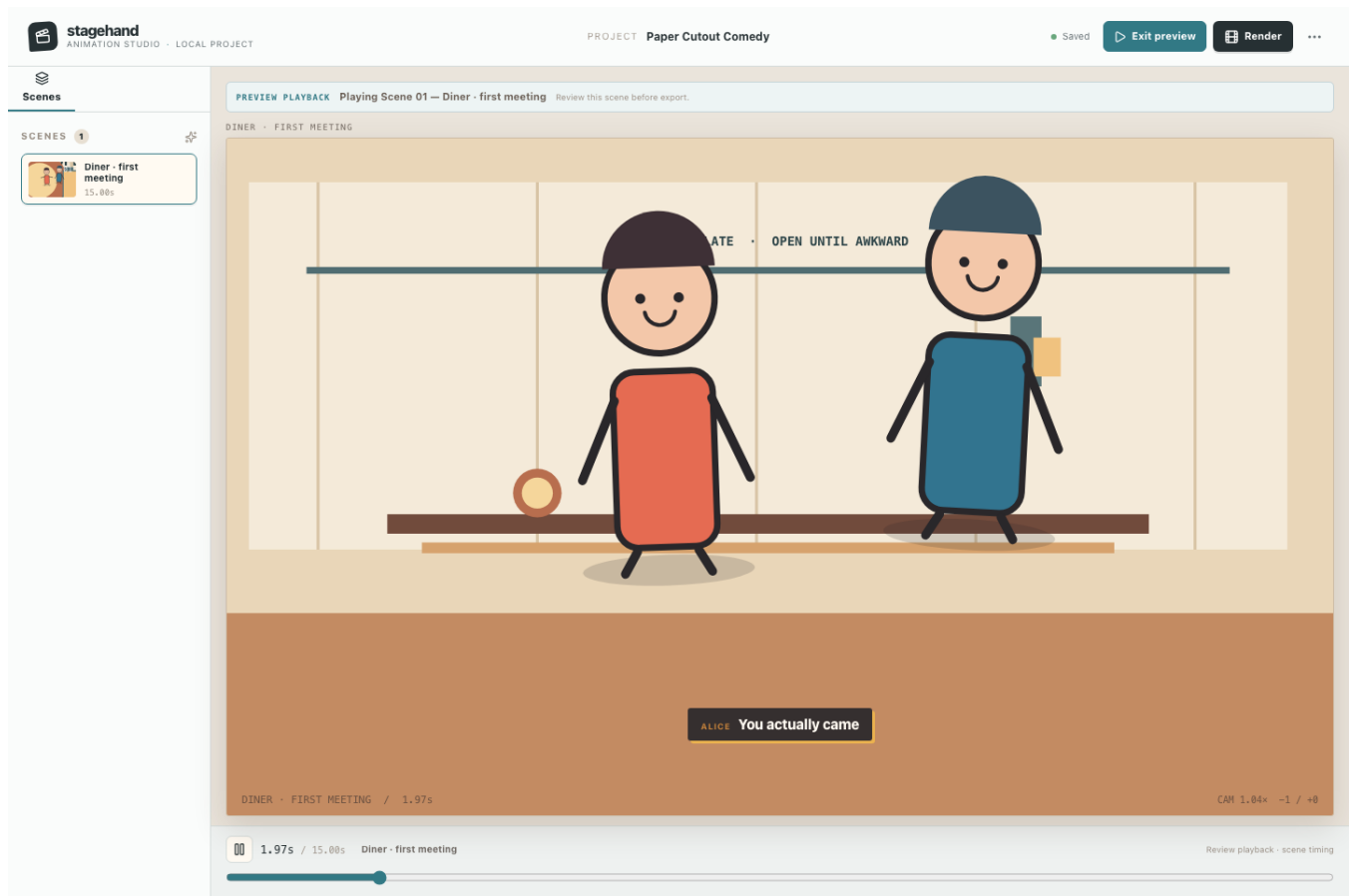
Hosted Playwright capture at 1440 x 960. Scannable asset rows stay in the rail while the selected asset's style and placement context live in Inspector.

Storyboard mode with Board rail



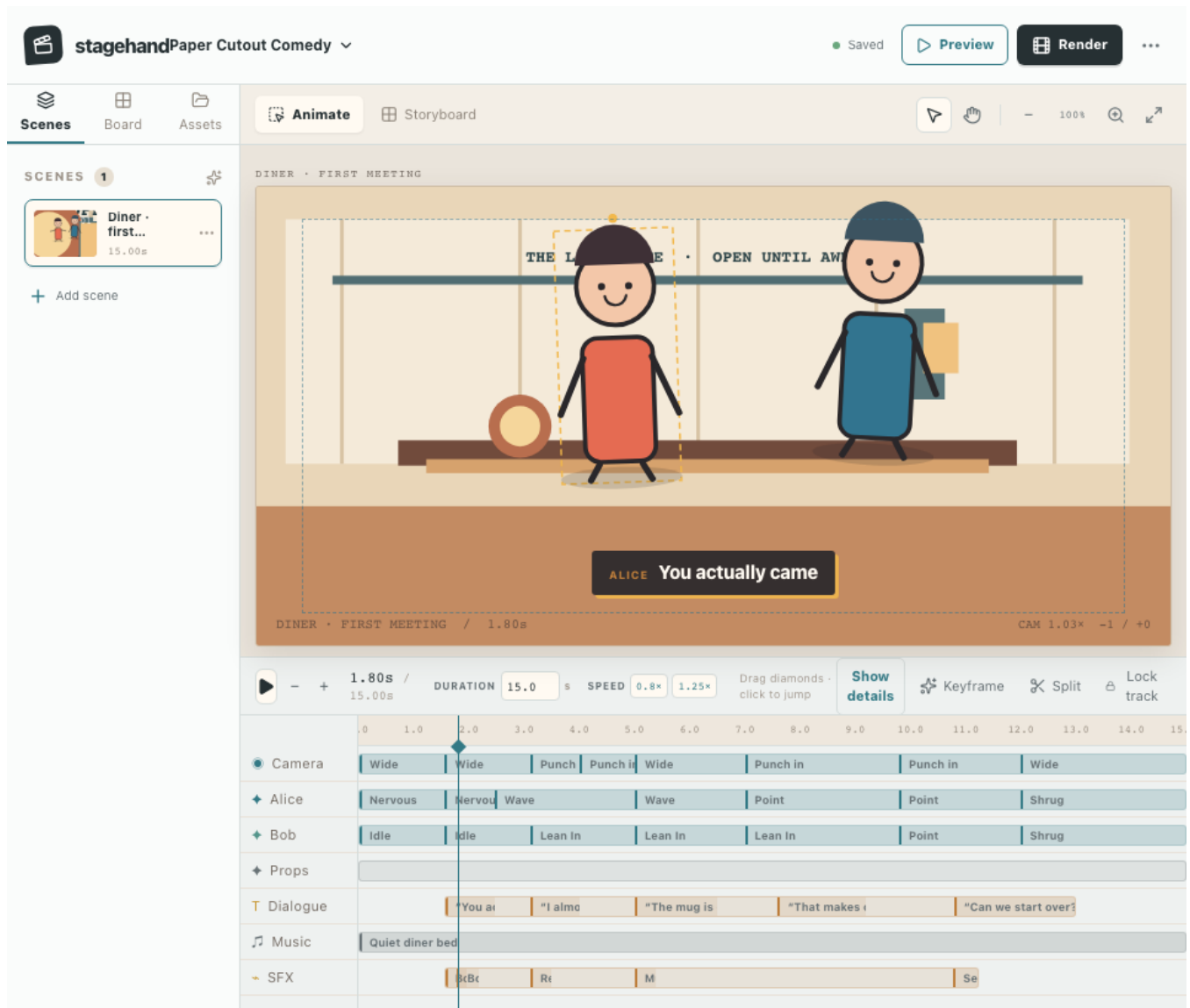
Hosted Playwright capture at 1440 x 960. Six beat cards make the longer arc inspectable while the selected mode and project rail stay synchronized.

Review-first Preview player



Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and a compact scrubber remain; the editable timeline grid and Inspector are hidden.

Split-browser compact editor



Hosted Playwright capture at 960 x 820. The Inspector becomes a closed-by-default drawer, leaving the scene rail and artwork usable beside a ChatGPT or Codex conversation.

Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields; later made those treatments visible in the canvas and manifest placement cues.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.
- Scene titles competed with the ready badge and were visually shortened. Used a two-line title clamp and positioned the badge outside the text flow.
- The smoke gate did not exercise recovery, modal semantics, or cross-scene playback. Added Help open/close, reload recovery, and sequence Preview transition checks to local and hosted runs.
- The Inspector was a long scroll dump, human scrubbing could leave agent reads one event behind, and keyboard stepping was unreliable after scrub focus. Added native collapsible groups, shared project-view synchronization, a focused stage target, and smoke coverage.
- Export checks only proved that downloads were non-empty. Added binary signature assertions for WebM/EBML and PNG outputs in local and hosted smoke runs.
- Experimental Chromium exposed navigator.modelContext while the app only checked document.modelContext. Added a compatibility fallback and a native registration smoke gate covering all 52 tools plus valid and invalid calls.
- Preview still read like the editor because raw timeline, guides, and status toast survived into review. Added a compact scene/timing scrubber and hid editing-only surfaces.

- The asset rail carried too much editing UI. Kept it scannable and moved the full style/brief editor into the wider Inspector.
- Header/sidebar implementation details competed with the artwork. Made Preview and Render primary actions, moved secondary actions under More, and removed permanent validation/tool-count/starter cards.
- The timeline's first read was keyframe noise. Added semantic event bands with an explicit Show details disclosure for precise edits.
- The desktop shell could force phone-width overflow. Removed the global minimum width and added responsive project/Inspector drawers with a readable-width timeline.
- In a split ChatGPT/Codex browser pane, the permanent Inspector consumed the stage's working width. Moved it to a closed-by-default overlay drawer at ≤ 1200 px, hid the redundant project-drawer action while the rail remains visible, and added a checked-in 960 px regression gate.

Limits

The regular hosted interaction suite remains a Playwright fallback, while a Sol-powered Codex run directly exercised the public page's WebMCP tools in the in-app browser and verified the Alice/Bob coupon gag. The Luna-bound main task cannot call browser WebMCP; supported Sol/Terra tasks can. No separate MCP server connection, TOML entry, or restart was needed. Native Chromium smoke complements that live run. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.